Quiz — 1.3.2 Databases

Name: ______________________ Date: __________ Mark: ______ / 25

OCR H446 · Component 01 · 1.3.2

Section A — Multiple choice (8 marks)

1. Compared with a flat file, a relational database typically has: A. More data redundancy B. Less data redundancy and fewer anomalies C. Only one table D. No primary keys

Your answer: ☐

2. A foreign key is: A. An attribute used only for sorting B. The primary key of another table, used to link tables C. A key made of two attributes D. A key that must always be null

Your answer: ☐

3. A composite primary key is one that: A. Links to another database B. Is made up of two or more attributes combined C. Is automatically generated D. Cannot be used to identify records

Your answer: ☐

4. A many-to-many (M:N) relationship is normally resolved by: A. Deleting one of the tables B. Adding a linking/junction table C. Removing the primary keys D. Using a secondary key only

Your answer: ☐

5. Referential integrity ensures that: A. Every foreign key references a valid existing primary key (or is null) B. All data is encrypted C. Tables can never be joined D. Records are always sorted

Your answer: ☐

6. A table is in 2NF if it is in 1NF and: A. Has no transitive dependencies B. Has no partial dependencies on a composite key C. Has no primary key D. Contains repeating groups

Your answer: ☐

7. Which ACID property guarantees that a transaction is "all or nothing"? A. Consistency B. Isolation C. Atomicity D. Durability

Your answer: ☐

8. Record locking is used during a transaction to: A. Speed up SELECT queries B. Prevent the lost update problem from concurrent edits C. Encrypt the record D. Remove transitive dependencies

Your answer: ☐

Section B — Short answer (12 marks)

9. State two differences between a flat file database and a relational database. **[2]**

10. Explain the rule a table must meet to be in 1NF. **[2]**

11. A table is in 2NF but not 3NF. Explain what must be removed to reach 3NF. **[2]**

12. Consider a table `Student(StudentID, Surname, Form, TutorName)`. Write an SQL query that returns the `StudentID` and `Surname` of every student in form `'12A'`, sorted by `Surname` in ascending order. **[3]**

13. Explain what is meant by the Isolation property of a transaction, and why it matters. **[2]**

14. State one benefit and one risk of using record locking. **[1]**

Section C — Exam-style question (5 marks)

15. A library uses a relational database with two tables:

`Member(MemberID, Name, Email)` `Loan(LoanID, MemberID, BookTitle, DueDate)`

(a) Identify the foreign key and the table it links to. **[1]**

(b) Write an SQL statement that joins the two tables to list each member's `Name` together with the `BookTitle` of every book they currently have on loan. **[3]**

(c) Explain how referential integrity prevents an invalid loan record. **[1]**